

Soham Sarkar

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Summary

Ph.D. candidate in Applied Mathematics with expertise in **numerical modeling**, **optimization**, and **machine learning**. Proficient in **MATLAB**, **Python** and **HPC**, with research focused on **stability and dynamics** of nonlinear dynamical systems.

Education

The University of Texas at Dallas

TX, United States

DOCTORATE, MATHEMATICS (APPLIED)

2022 - Present

- Advisors - John Zweck and Dmitry Rachinskiy

Savitribai Phule Pune University

Pune, India

MASTER OF SCIENCE, MATHEMATICS

2018 - 2020

- Specialization - Commutative algebra

University of Calcutta

Kolkata, India

BACHELOR OF SCIENCES (HONOURS), MATHEMATICS

2014 - 2018

- Minors - Physics and Chemistry

Relevant Coursework

Numerical PDEs, Numerical Linear Algebra, Optimization, Machine Learning, Statistical Modeling, Scientific Computing (Fluid flows), Data Management

Technical Skills

Programming MATLAB, Python, R, C, OpenMPI

Libraries NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Scanpy, caret, dplyr, tidyr, ggplot2

Data Management Hadoop DFS (MapReduce), Spark SQL

Typesetting software L^AT_EX, R Markdown, MS Office, GitHub

Professional Experience

The University of Texas at Dallas

Aug, 2022 - Dec, 2023; May, 2024 -

Present

GRADUATE TEACHING ASSISTANT

- **Numerical Analysis** (MATH 4334) - Spring '26, Fall '24, Spring '25.
- **Mathematical Analysis II** (MATH 4302) - Spring '26, '25.
- **Calculus-I** (MATH 2417) - Fall '25, Summer '23
- **Mathematical Analysis I** (MATH 4301) - Summer '25.
- **Elementary Analysis II** (MATH 5302) - Spring '25 - **Graduate Course**.
- **Differential Equations with Applications** (MATH 2420) - Fall '24
- **Linear Algebra** (MATH 2418) - Summer '24
- **Calculus of Several Variables** (MATH 2415) - Active learning model with assistance from ULAs - Fall '23
- **Integral Calculus** (MATH 2414) - Spring '23
- **Applied Calculus-II** (MATH 1326)-Spring '23.
- **Differential Calculus** (MATH 2413) - Fall '22.
- **Applied Calculus-I** (MATH 1325)-Fall '22

Jan - May, 2024

GRADUATE RESEARCH ASSISTANT (PI: DR. JOHN ZWECK)

- Devised a **two-level model** of SESAM in order to model realistic loss (and, gain) with pulse saturation.
- Determined **stability by linearizing pulse output** in SESAM component of a lumped laser model by Kim.

Chegg Inc.

Feb, 2021 - July, 2022

SUBJECT MATTER EXPERT - ADVANCED MATH

- Delivered expert solutions in Advanced Mathematics on Chegg, leveraging quantitative reasoning and analytical problem-solving to address diverse, complex problems spanning **applied mathematics** and **engineering**.

Research Experience

Stability of short pulse lasers with realistic saturable loss and amplification

Aug, 2023 - Present

ADVISOR: PROF. JOHN ZWECK, THE UNIVERSITY OF TEXAS AT DALLAS

- Developed **MATLAB-based optimization model** in lumped lasers with SESAM **with order 1 accuracy**.
- Linearized the pulse to determine its stability using **finite difference scheme** by computing its essential spectrum.
- Implemented adjoint-state method to incorporate **gradient-based BFGS optimization** in order to find the optimal pulse.

Zero Divisor Graphs of Commutative Rings and Posets

Feb - Aug, 2021

SUPERVISOR: PROF. VINAYAK JOSHI, SAVITRIBAI PHULE PUNE UNIVERSITY

- Studied a subclass of counterexamples to **Beck's coloring conjecture** on zero divisor graphs.
- Associated line graph characterization of subgroup lattices of cyclic groups.
- Defined **graph isomorphism** between comaximal subgroup graph lattice of a group and the zero divisor graph of its dual lattice.

Molien's Theorem to compute Invariants

Jan - Apr, 2020

SUPERVISOR: PROF. GANESH S.KADU, SAVITRIBAI PHULE PUNE UNIVERSITY

(Project report available [here](#))

- Formed unitary representation of group on symmetric algebra of dual vector space and its unitary induction on graded algebras.
- Defined **spectrum of Reynold's operator** and certain properties of unitary operators.
- Defined grading on algebra of invariants and formulated the series.
- Authored **highest graded** Master's research project (8/10).

Expander Graph and its Applications

Jan - Apr, 2020

SUPERVISOR: PROF. GANESH S.KADU, SAVITRIBAI PHULE PUNE UNIVERSITY

(Project report available [here](#))

- Proved Expander-mixing lemma on regular **random graphs**.
- Explored the construction and properties of one-sided expander graphs.

Talks and Presentations

GRADUATE SPEAKER

- Poster presentation on "**Stability of pulses in a lumped model of a laser with a slow saturable absorber**" at UT Austin (SIAM TX-LA Sectional Meeting, 2025).
- Presentation on "Heart failure prediction using multiple **linear and logistic regression models**" at UT Dallas (Spring 2025).
- Lightning talk on "Stability in a short pulse laser" at "Dynamical Systems and Fractal Geometry" conference, UNT Denton from May 14 to 17, 2024.
- Thesis Proposal Exam (ORCE) on **Stability of short pulse lasers with realistic saturable loss and amplification** for Advisory Committee on 6th May, 2024.
- Presented the chapters on **Probability and Information Theory** from the book Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville weekly from September 26 to October 17, 2023 (Fall 2023).
- Presented the classical paper of Mallat's Theory on **multi-resolution signal decomposition** in terms of wavelets and its application into computer vision, pattern recognition and signal analysis on 18th October, 2022 (Fall 2022).

Certifications

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| 2021 | Using Python to Access Web Data , University of Michigan, COURSERA |
| 2021 | Python Data Structures , University of Michigan, COURSERA |
| 2021 | Programming for Everybody , University of Michigan, COURSERA |
| 2020 | German - A1 , IIT Madras, NPTEL |

Leadership roles and Extracurriculars

Communication and Outreach Coordinator

Sep, 2024 - Sep, 2025

SIAM UTD-SMU STUDENT CHAPTER

- Organized **Computational Sciences seminars**, **career panels** and **industry talks** in collaboration with SIAM chapter presidents.

EXTRACURRICULARS

- Hiking and endurance training.
- Active participant in marathon events.
- Vocalist at inter-school competitions.
- Former member of student committee of UDAAN - Cultural society at University of Calcutta.